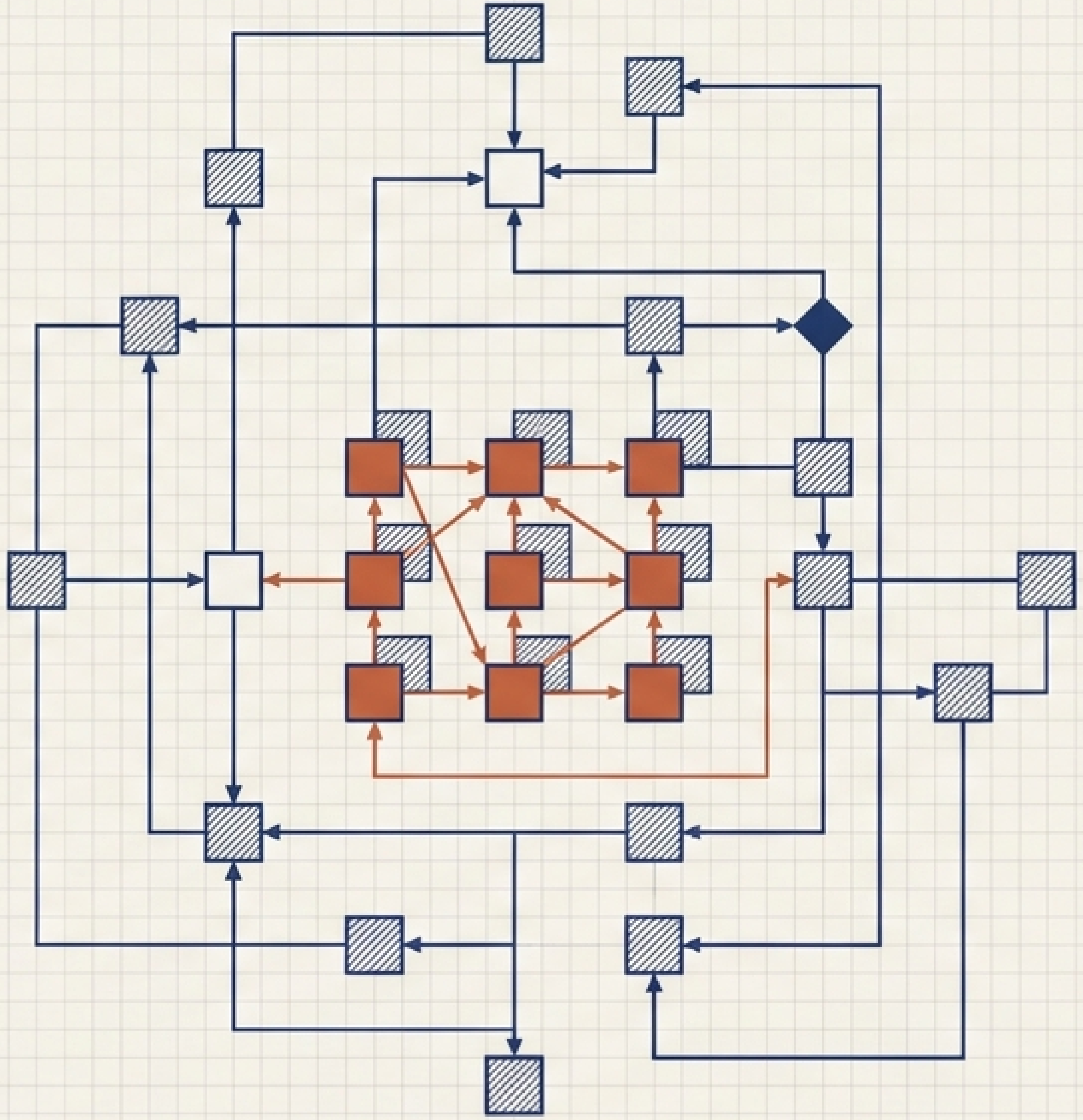
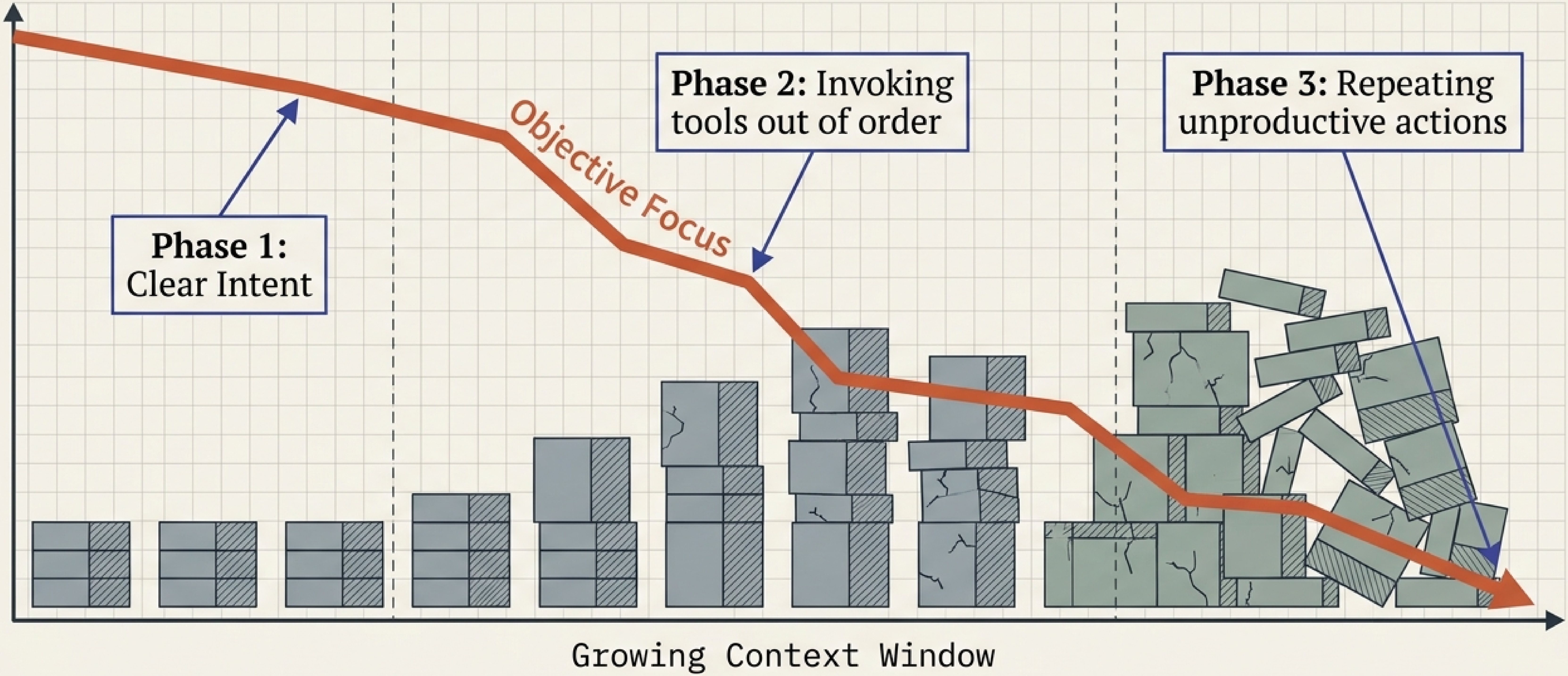
Procedural Graphs

Self-Evolving Execution Structures for LLM Agents

Based on research by Yuxing Lu, Yicheng Chen, Shanchan Wu, and Sercan Ö. Arık (Google, Georgia Tech, Peking University)

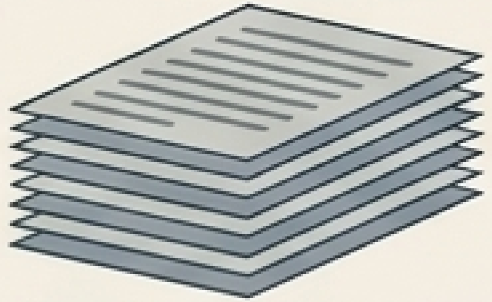
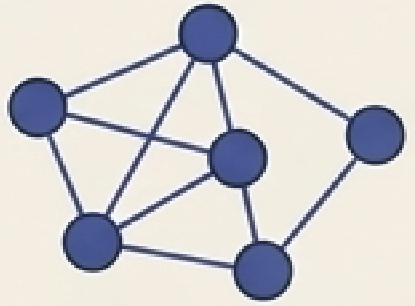
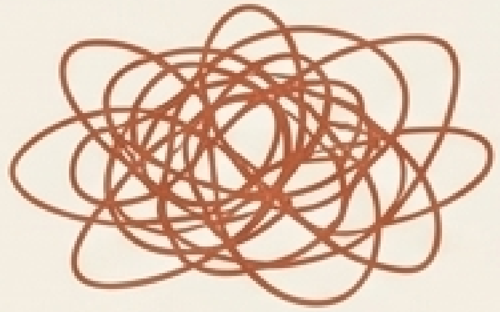
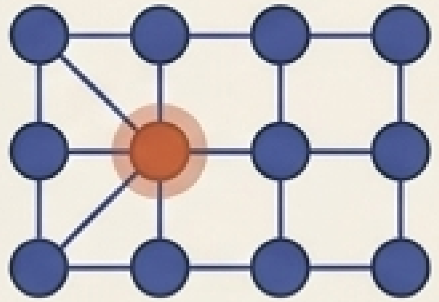


Timeline Degradation Chart



Unconstrained generation over an accumulating history leaves procedural knowledge implicit. As trajectories lengthen, agents simply lose track of what to do.

Memory Architecture Matrix

	Accumulating History (Implicit)	Procedural Graphs (Explicit)
Form of Knowledge	Implicit inside a growing text log. 	Explicit structured pathways. 
Navigation Method	Unconstrained generation. 	Localized graph traversal. 
Handling Long Horizons	Rapid degradation of focus.	Stable, step-by-step situational awareness.
Correction Mechanism	Manual prompt engineering.	Self-evolving topological edits.

The Triplet Evolution

Knowledge Graph

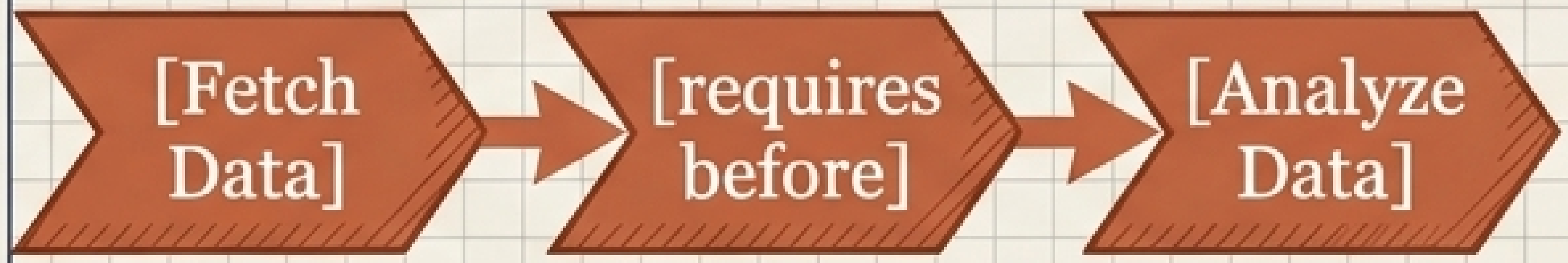
Answers: What is

(Entity) → Relation → (Entity)

[Paris] → is capital of → [France]

Procedural Graph

Answers: What to do



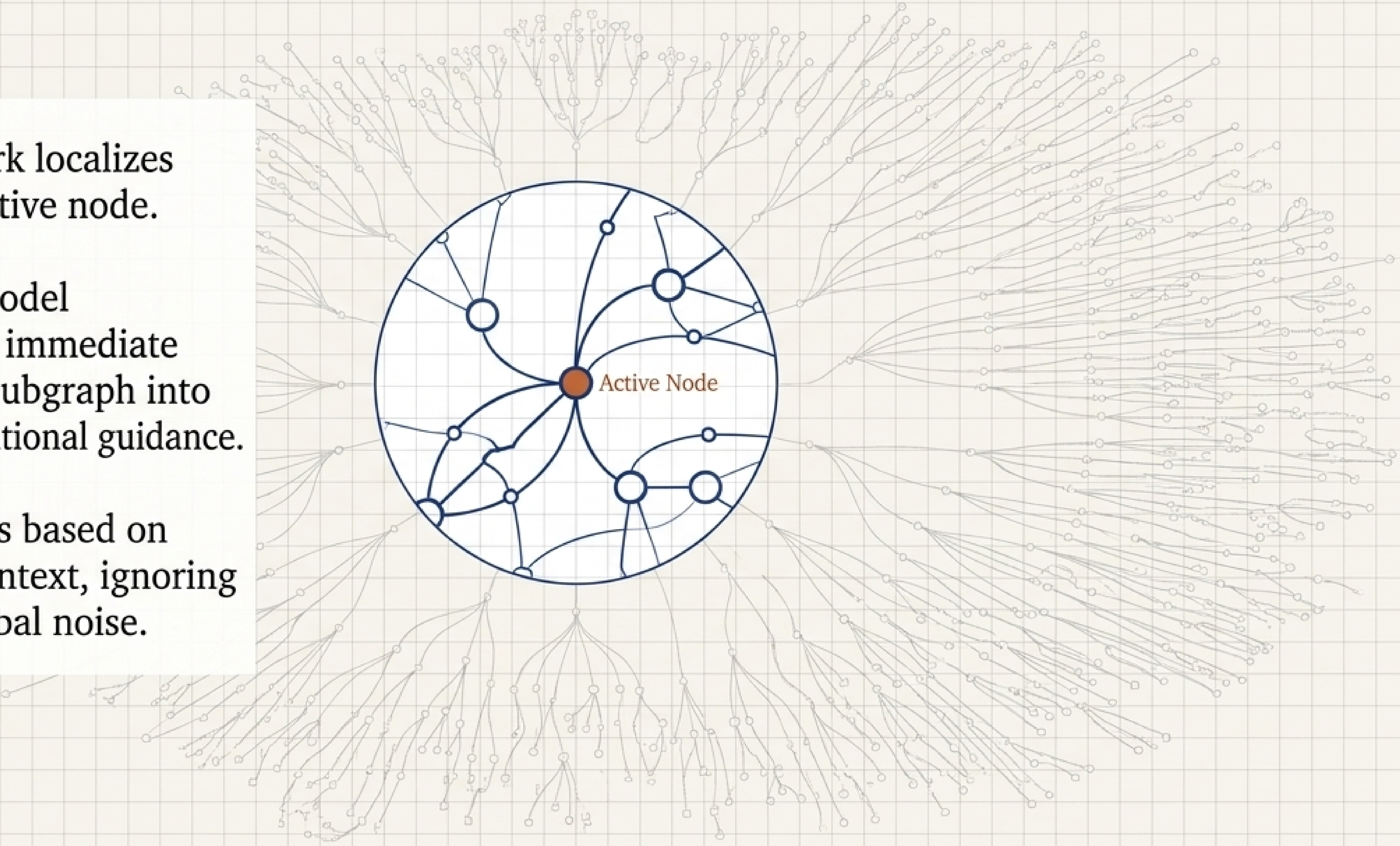
(Procedure) → Relation → (Procedure)

[Fetch Data] → requires before → [Analyze Data]

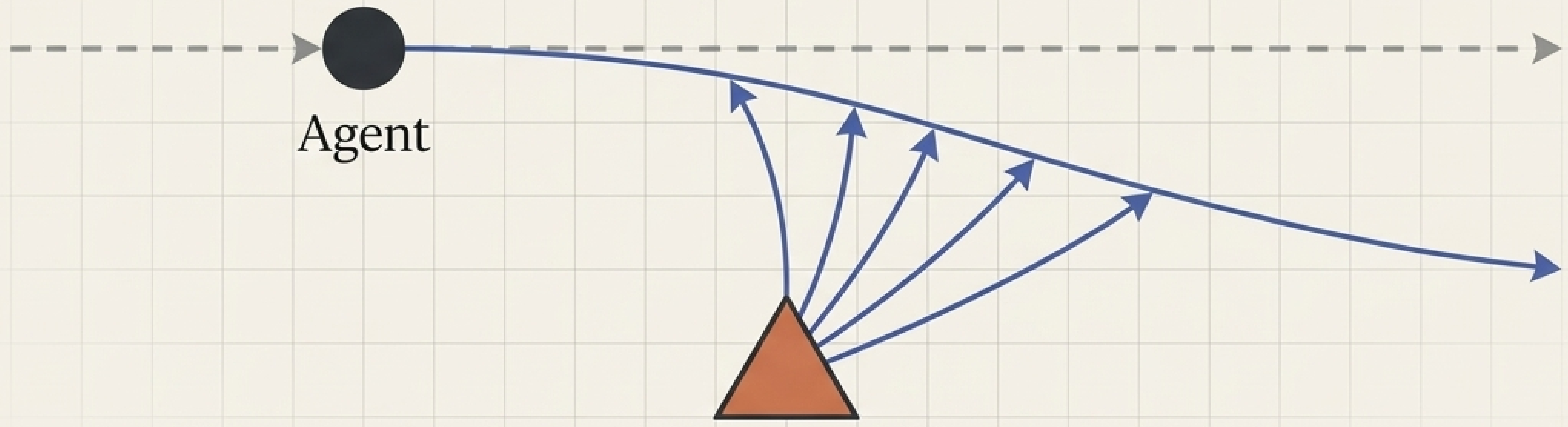
Instead of organizing factual knowledge, Procedural Graphs organize execution logic into a queryable structure.

Procedural Graph Execution Flow

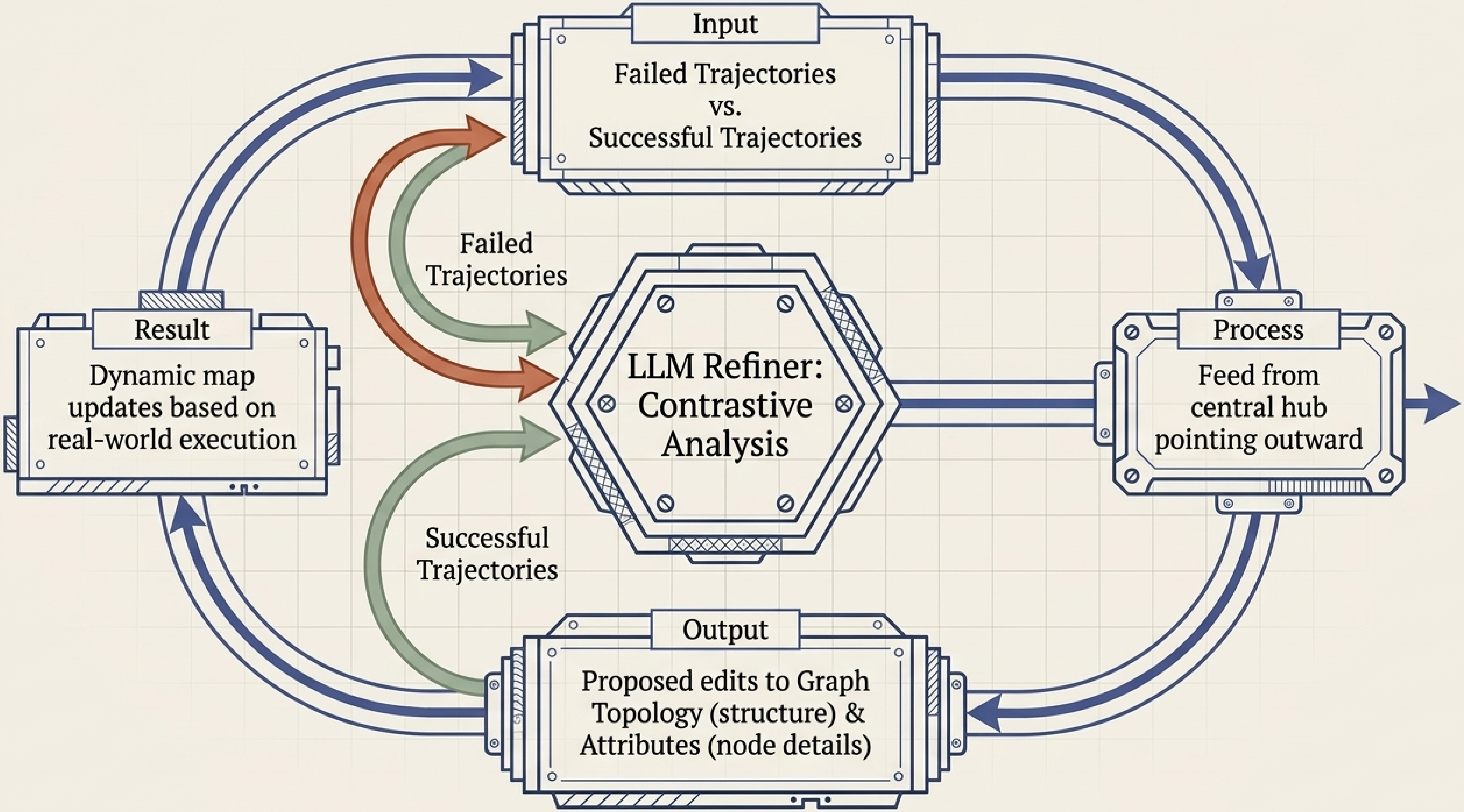
1. The framework localizes the agent's active node.
2. A guidance model translates the immediate surrounding subgraph into step-level situational guidance.
3. The agent acts based on immediate context, ignoring irrelevant global noise.



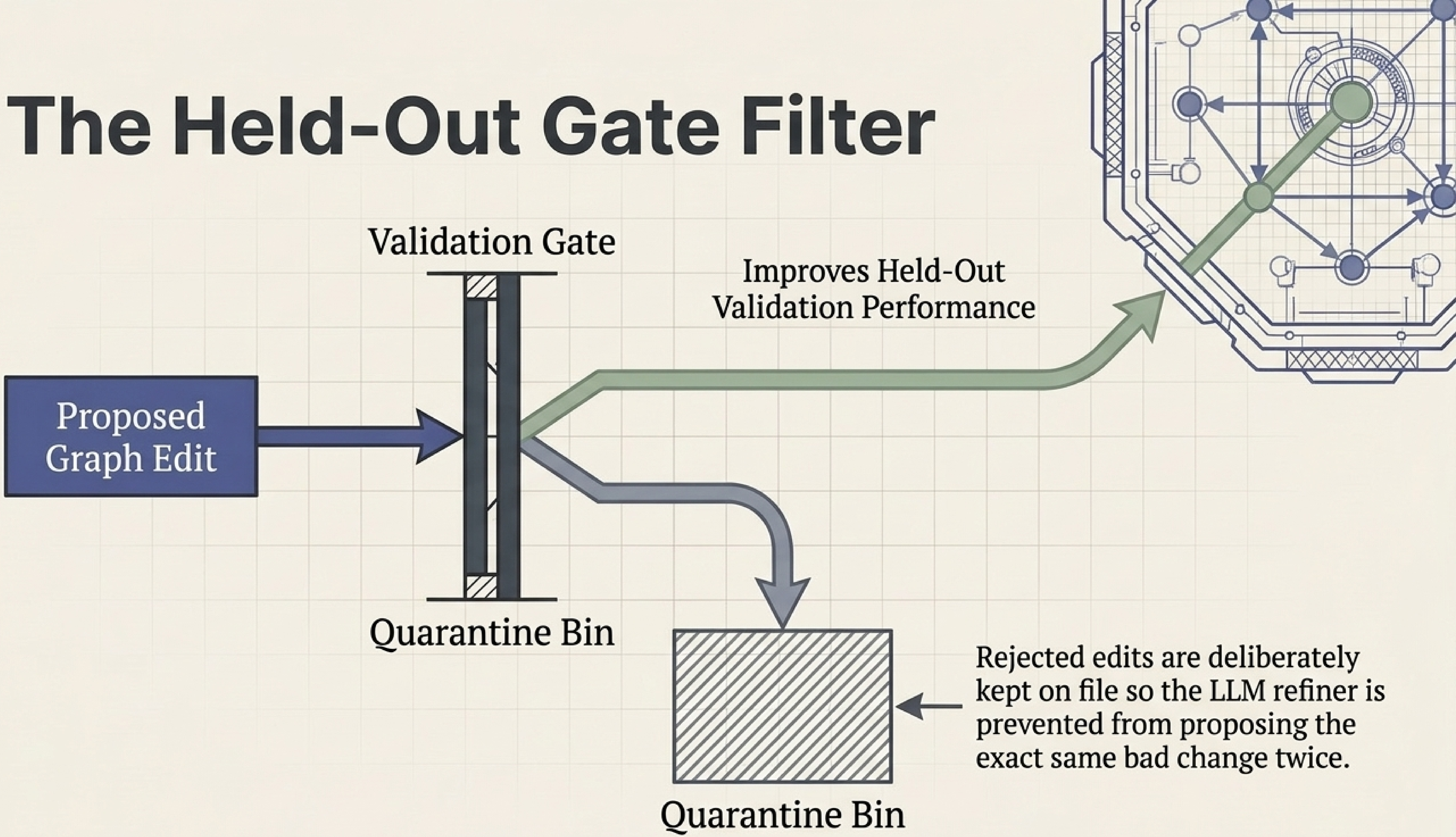
Guidance is a bias, not a command.



The procedural graph shifts the solver's next action without dictating it. If the graph's suggestion is wrong, the solver retains enough autonomy to deviate and break orbit.

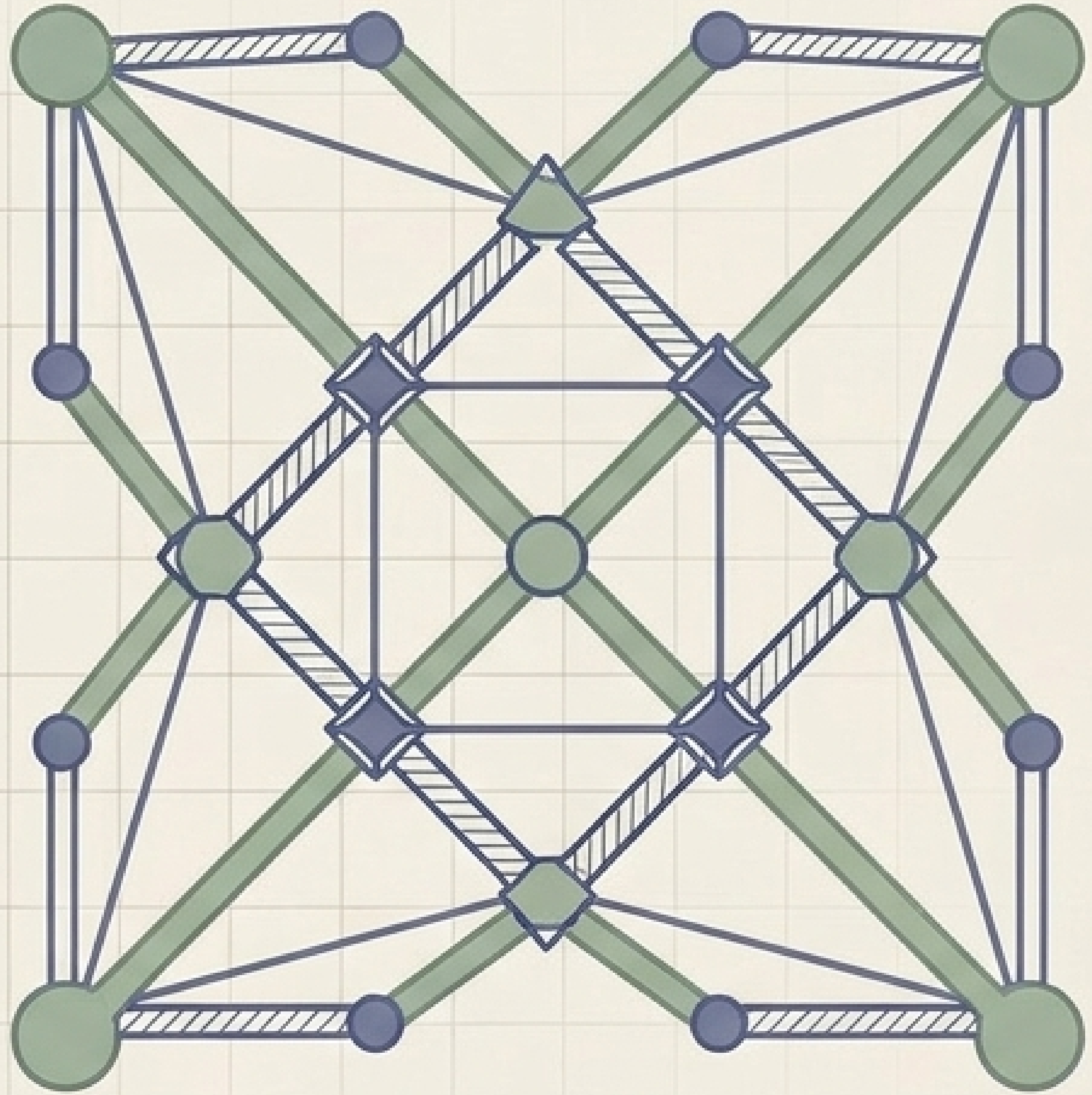

Input
Failed Trajectories
vs.
Successful Trajectories
Failed
Trajectories
LLM Refiner:
Contrastive
Analysis
Result
Dynamic map
updates based on
real-world execution
Process
Feed from
central hub
pointing outward
Successful
Trajectories
Output
Proposed edits to Graph
Topology (structure) &
Attributes (node details)
Gemini Notebook

The Held-Out Gate Filter



Flawed Expert Prior

Optimized Agentic Structure



The system can start from a minimal skeleton or a deeply flawed human design, and autonomously evolve a graph that matches or surpasses hand-designed alternatives.

Delivers consistent gains over standard memory-based baselines.

Performance scales across multiple distinct datasets and task types.

Agnostic to the backbone: functions effectively across various LLMs.

Self-evolution drives continuous improvement with zero manual engineering.

An Immune System for Agentic Workflows.

True agentic autonomy does not come from infinitely larger context windows. It requires an explicit, structured procedural memory that guides execution and actively inoculates itself against repeated mistakes.